

MA.912.AR.9.6

 Benchmark

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

Benchmark Clarifications:

Clarification 1: Instruction focuses on analyzing a given function that models a real-world situation and writing constraints that are represented as linear equations or linear inequalities.

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.2.2, MA.912.AR.2.5, MA.912.AR.2.7, MA.912.AR.2.8
- MA.912.AR.3.8

Terms from the K-12 Glossary

- Inequality
- Linear Equation

Vertical Alignment

Previous Benchmarks

- MA.8.AR.2.2
- MA.8.AR.4

Next Benchmarks

- MA.912.AR.4.4
- MA.912.AR.5.7, MA.912.AR.5.9
- MA.912.AR.6.6
- MA.912.AR.7.3, MA.912.AR.7.4
- MA.912.AR.8.3
- MA.912.AR.9.7, MA.912.AR.9.10
- MA.912.T.3.3

Purpose and Instructional Strategies

In grade 8, students worked with linear equations and inequalities, and graphically solved systems of linear equations. In Algebra I, students represent constraints as systems of linear equations or inequalities and interpret solutions as viable or non-viable options. In later courses, students will solve problems involving linear programming and work with constraints within various function types.

- For students to have full understanding of systems, instruction includes MA.912.AR.9.4 and MA.912.AR.9.6. Equations and inequalities and their constraints are all related and the connections between them should be reinforced throughout the instruction.
- Allow for both inequalities and equations as constraints. Include cases where students must determine a valid model of a function.
 - Students often use inequalities to represent constraints throughout Algebra I. Equations can be thought of as constraints as well. Solving a systems of equations requires students to find a point that is constrained to lie on specific lines simultaneously.
- Instruction includes the use of various forms of linear equations and inequalities.
 - Standard Form
Can be described by the equation $Ax + By = C$, where A , B and C are any rational number and any equal or inequality symbol can be used.
 - Slope-intercept form
Can be described by the inequality $y \geq mx + b$, where m is the slope and b is the y -intercept and any equal or inequality symbol can be used.
 - Point-slope form
Can be described by the inequality $y - y_1 > m(x - x_1)$, where (x_1, y_1) are a point on the line and m is the slope of the line and any equal or inequality symbol can be used.

Common Misconceptions or Errors

- Students may have difficulty translating word problems into systems of equations and inequalities.
- Students may shade the wrong half-plane for an inequality.
- Students may graph an incorrect boundary line (dashed versus solid) due to incorrect translation of the word problem.
- Students may not identify the restrictions on the domain and range of the graphs in a system of equations based on the context of the situation.

Strategies to Support Tiered Instruction

- Instruction provides opportunities to translate systems of equations or inequalities from word problems by first creating equations, then by identifying key words to determine the inequality symbol (i.e., no more than, less than, at least, etc.). The appropriate inequality symbols can then replace the equal signs. Students can separate and organize information for each equation or inequality.
 - Separate given information for each equation or inequality
 - Determine the appropriate form of equation or inequality based on givens
 - Define a variable to represent the item wanted in the equation or inequality
 - Determine what values are constants or should be placed with the variables
 - Write the equation or inequality
- Teacher makes connections back to students' understanding of MA.912.AR.2.5 and MA.912.AR.3.8 and writing constraints based on a real-world context.
 - For example, Dani is planning her wedding and the venue charges a flat rate of \$8250 for four hours. The venue can provide meals for each of the guests and charges \$21.25 per plate for adults and \$13.75 per plate for children if she has a minimum of 75 guests. If Dani's budget is \$38,000, students can describe this situation using the inequalities $a + c \geq 75$ and $21.25a + 13.75c + 8250 \leq 38000$. Depending on number of adults and children she wants to invite and the capacity of the venue, students can determine various other constraints.
- Teacher provides questions to be answered by students to aid in the identification of domain and range restrictions:
 - Does the problem involve humans, animals, or things that cannot or normally not be broken into parts? If yes, you are restricted by integers.
 - Do negative numbers not make sense? If yes, you are restricted by positive numbers.
 - Was a maximum or minimum value given? If yes, the solution must not exceed the maximum or drop lower than the minimum.
- Instruction includes identifying which variable(s) the constraints apply to.

Instructional Tasks

Instructional Task 1 (MTR.3.1)

A baker has 16 eggs and 15 cups of flour. One batch of chocolate chip cookies requires 4 eggs and 3 cups of flour. One batch of oatmeal raisin cookies requires 2 eggs and 3 cups of flour. The baker makes \$3 profit for each batch of chocolate chip cookies and \$2 profit for each batch of oatmeal raisin cookies. How many batches of each cookie should she make to maximize profit?

Instructional Task 2 (MTR.4.1)

Amy and Anthony are starting a pet sitter business. To make sure they have enough time to properly care for the animals they create a feeding and pampering plan. Anthony can spend up to 8 hours a day taking care of the feeding and cleaning and Amy can spend up to 8 hours each day on pampering the pets.

Feeding/Cleaning Time: Amy and Anthony estimate they need to allot 6 minutes twice a day, morning and evening, to feed and clean litter boxes for each cat, a total of 12 minutes a day per cat. Dogs will require 10 minutes twice a day to feed and walk, for a total of 20 minutes per day for each dog.

Pampering Time: Sixteen minutes per day will be allotted for brushing and petting each cat and 20 minutes each day for bathing and playing with each dog.

Part A. Write an inequality for feeding/cleaning time needed for the pets. Represent all time in the same unit (minutes or hours).

Part B. Write an inequality for pampering time needed for the pets. Represent all time in the same unit (minutes or hours).

Part C. Graph the two inequalities.

Part D. In term of this scenario, explain the meanings of the following points: (0,24) and (30,0).

Part E. What is the greatest number of dogs they can watch if they are watching 19 cats?

Part F. List two viable combinations of pets that can be watched.

Possibility 1: _____ cats _____ dogs

Possibility 2: _____ cats _____ dogs

Instructional Items

Instructional Item 1

There are several elevators in the Sandy Beach Hotel. Each elevator can hold at most 12 people. Additionally, each elevator can only carry 1600 pounds of people and baggage for safety reasons. Assume on average an adult weighs 175 pounds and a child weighs 70 pounds. Also assume each group will have 150 pounds of baggage plus 10 additional pounds of personal items per person.

Part A. Write a system of linear equations or inequalities that describes the weight limit for one group of adults and children on a Sandy Beach Hotel elevator and that represents the total number of passengers in a Sandy Beach Hotel elevator.

Part B. Several groups of people want to share the same elevator. Group 1 has 4 adults and 3 children. Group 2 has 1 adult and 11 children. Group 3 has 9 adults. Which of the groups, if any, can safely travel in a Sandy Beach elevator?

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*